Endpoint Analytics Deployment & Optimization in Microsoft Intune — Startup, Reliability & Work-From-Anywhere Scores

Implementation Solution Guide

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Executive Summary — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Most endpoint teams have **no objective measure of the user experience**. They learn about slow boots, crashing apps, and struggling remote laptops the same way every time — when the tickets pile up. There is no baseline, no benchmark, and no way to prove whether last quarter's work actually made devices faster or more reliable.

Endpoint Analytics is the fix: the **data plane Microsoft Intune uses to measure the real endpoint experience** — Startup performance, Application + Device reliability, Work-From-Anywhere readiness, and Resource performance, all **scored and benchmarked**. This solution deploys it end to end and turns the data into action:

• **Turn it on, the right way** → enable Endpoint Analytics and assign a named **data-collection

policy **to a scoped** pilot** group, so telemetry starts flowing before you touch the whole fleet.

• **Baseline and benchmark** → establish a pilot baseline, then a fleet baseline, and compare against

the Microsoft-published "all organizations" median — so every later number means something.

• **Read every score, and the *why* under it** → interpret all four scores down to the sub-signals

(the Group-Policy processing time, the crashing app, the outdated driver, the legacy Windows edition) that explain a low number.

• **Ship a prioritized backlog** → convert the worst offenders into a ranked remediation plan, then run

a review cadence that re-measures against the baseline so improvement is provable.

Built the standard repeatable way — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with exact, click-by-click Intune steps, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph, read-only plus one `-WhatIf`-gated state-changer), and testing/troubleshooting guides with clear pass/fail gates.

This is the **flagship** of a three-product Endpoint Analytics set: it stands up, baselines, and operationalizes Endpoint Analytics, then hands the two deep specialties to its companions — **Proactive Remediations at Scale** to fix the backlog with detection/remediation script packs, and **Anomaly Detection & Application Reliability** for the reliability and anomaly deep-dive.

Outcome: an Endpoint Analytics deployment that gives you an objective, benchmarked view of the endpoint experience and a prioritized plan to improve it — managed by data instead of ticket volume, by your own team, without a consulting engagement.

1. Solution Overview — Endpoint Analytics Deployment & Optimization in Microsoft Intune

What this solution is

This solution turns on **Endpoint Analytics** — the **data plane Microsoft Intune uses to measure the real endpoint experience** — and puts it to work end to end. Instead of learning about slow boots, crashing

apps, and struggling remote laptops only when the help desk gets buried, you get an **objective, scored, benchmarked** view of how devices actually behave for the people using them, and a **prioritized plan to improve it**.

Endpoint Analytics collects experience telemetry from your managed Windows devices, rolls it up into a small set of **scores** (with the detailed **sub-signals** behind each), and compares those scores against a **baseline** — your own historical baseline and the Microsoft-published "all organizations" median. This package deploys that capability the standard, repeatable way — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — then teaches you to read each score, read the sub-signals under it, and convert the worst offenders into a **remediation backlog** you can work down.

It delivers exactly what an endpoint team needs to run experience by data:

1. **Endpoint Analytics enabled** on the tenant, with a named **data-collection policy** assigned to a scoped device group so telemetry starts flowing.
1. **A baseline** — a pilot baseline first, then a fleet baseline — so every later number has something to be measured against.
1. **Every score interpreted** — Startup performance, Application + Device reliability, Work-From-Anywhere, and Resource performance — down to the sub-signals that explain *why* a score is low.
1. **A remediation backlog** — the low scores and their worst-contributing devices, apps, and drivers turned into a ranked, actionable list, plus a repeatable **review cadence** to keep improving.

The four score areas (and what each measures)

Endpoint Analytics presents its data as a set of scores. Cover all four at a working level; the exact score names, weightings, thresholds, and blades change over time, so treat the specifics as **verify-against-current-Microsoft-documentation** and the concepts as durable.

• **Startup performance.** How long a device takes to become usable from power-on through to a responsive desktop — broken into phases (core boot, sign-in, and the time to a responsive desktop). The sub-signals point at what is slow: **Group Policy / configuration processing**, heavy **startup apps**, **disk type** (spinning vs. SSD), and outdated **drivers/firmware**. This is the flagship score most teams start with because slow logon is the most-felt pain.

• **Application reliability.** How reliably the apps people actually run stay up — surfacing apps with high **crash/hang rates**, the devices they crash on, and usage counts so you can tell a broadly painful app from a one-device fluke. *The deep reliability + anomaly analysis is the subject of the companion product — see "Where this fits" below.*

• **Work-From-Anywhere (WFA).** How ready your fleet is to be managed and secured from anywhere — a readiness roll-up over **cloud identity** (Entra-joined), **cloud management** (Intune/co-managed), **Windows edition/version currency**, **Autopilot** provisioning, and **TPM/hardware** readiness. It is less "how fast" and more "how modern and remote-ready."

• **Resource performance.** How hard devices are working — a spotlight on **CPU** and **RAM**

pressure so you can spot under-spec'd or over-loaded machines that make the experience sluggish even when boot times look fine.

Score names, exact sub-metrics, and thresholds evolve; confirm the current set in the Intune portal and Microsoft documentation before quoting a specific number to stakeholders.

How baselining and benchmarking work

A raw score means little on its own — the value is in the comparison:

1. **Data collection starts the clock.** Once the data-collection policy is assigned and devices are reporting, Endpoint Analytics needs a **collection window** before scores populate (they do not appear instantly — see Troubleshooting). Confirm exact timing against current Microsoft docs.

1. **A baseline is captured.** Endpoint Analytics keeps a **baseline** you compare current scores against. You start with a **pilot baseline** (a small scoped group), then a **fleet baseline** once the whole managed estate is reporting. You can also compare against the **Microsoft-published "all organizations" median baseline** to see whether you are ahead of or behind the pack.

1. **Sub-signals explain the gap.** When a score sits below baseline, you drill into its sub-signals to find the *cause* — e.g. a low Startup score that is really a Group-Policy-processing problem, or a WFA score dragged down by devices still on a legacy Windows edition.

1. **Improvement is measured against the baseline.** After you act on the backlog, the score should move toward (and past) baseline — that delta is your proof the work paid off.

The improvement loop

This is the operating rhythm the solution installs:

> ***Baseline → interpret each score → drill to sub-signals → build a prioritized remediation backlog → remediate → re-measure against the baseline → repeat on a review cadence.***

Two of those steps have **dedicated companion products** so this flagship stays focused on deploying, baselining, and interpreting rather than re-teaching the deep specialties:

Where this fits — the Endpoint Analytics cluster (pointers, not repetition)

This is the **flagship** of a three-product Endpoint Analytics cluster. It covers all four scores at a working level and hands off the two deep specialties to its siblings — **use those for depth; this guide does not reproduce them**:

- **Proactive Remediations at Scale — Detection & Remediation Script Packs for Microsoft Intune (#2).**

Endpoint Analytics surfaces *what* to fix; **Proactive Remediations** is the **engine that fixes it at scale** — authoring, packaging, scoping, and scheduling the detection/remediation script pairs that clear items off your backlog automatically. When this guide says "remediate the backlog item," the *how* of building the script pack lives there.

- **Anomaly Detection & Application Reliability with Endpoint Analytics (#3).** This flagship reads the

Application/Device reliability score at a working level; the **reliability and anomaly deep-dive** — interpreting crash/hang patterns, using anomaly detection to catch a bad app/update/driver rollout early, and running a reliability investigation to root cause — lives in that companion product.

Buy this one to **stand up, baseline, and operationalize** Endpoint Analytics; add #2 to **remediate at scale** and #3 to **go deep on reliability and anomalies**.

The standard deployment model

This solution uses **one** model so the process is repeatable across the catalog:

> ***Enable Endpoint Analytics on the tenant, assign a named data-collection policy to a scoped pilot group, let devices report and establish a baseline, interpret each score and its sub-signals, turn the low scores into a prioritized remediation backlog, then expand from pilot to the whole fleet — improving on a review cadence.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Benefits

- **Experience becomes measurable** — objective, scored, benchmarked numbers replace "it feels slow"

and ticket volume as your signal.

- **You know *why*, not just *what*** — the sub-signals under each score point at the actual lever

(policy, app, driver, disk, hardware, edition) so remediation is targeted.

- **A prioritized backlog, not a data dump** — the worst-scoring devices/apps/drivers become a ranked

list your team can work down.

- **Proof of improvement** — every change is re-measured against the baseline, so you can show

stakeholders the score moved.

- **Fleet-scale and repeatable** — assign the data-collection policy once to a group; add devices to

the group and they start contributing. The same playbook runs on the next wave and the next project.

- **Composable with the cluster** — pairs with Proactive Remediations (#2) to fix at scale and with

the reliability/anomaly deep-dive (#3) without re-buying overlapping content.

Limitations (be honest with buyers)

- **Windows-focused.** Endpoint Analytics scores are centered on **managed Windows devices**; it is not

a cross-platform experience score for iOS/Android/macOS. Confirm current platform support against Microsoft documentation.

- **Scores are not instant.** After you enable data collection there is a **collection window** before

scores populate and a baseline forms — plan for it; don't declare failure on day one.

- **Licensing and feature availability can vary.** Endpoint Analytics availability and the specific

features/scores you see depend on your licensing and tenant configuration — **verify against current Microsoft documentation** for your plan, and note that **national clouds (GCC, GCC High, DoD)** can differ in feature parity, rollout timing, and portal endpoints.

- **It measures and prioritizes; it does not fix by itself.** Turning a score up requires action —

policy changes, driver updates, hardware refresh, or **Proactive Remediations (#2)** script packs. This product builds the backlog; remediation-at-scale is the sibling's job.

- **Data collection depends on connectivity and endpoints.** Devices must be able to reach the required

Microsoft service endpoints to submit analytics data; blocked endpoints mean blank scores.

2. Requirements — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (tenant, Intune licensing, admin roles, supported OS, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to standing up **Endpoint Analytics**, getting devices to report telemetry, baselining, and interpreting the scores.

> ***Licensing, feature availability, and national-cloud specifics change.*** *Where this document names a license, a portal path, or a service endpoint, treat it as a starting point and* ***verify against current Microsoft documentation*** *for your plan and cloud (Commercial vs. GCC / GCC High / DoD) before you rely on it.*

R1. An Endpoint Analytics–capable license

Description: A Microsoft license/plan that includes **Endpoint Analytics** for your managed devices. **Why:** Endpoint Analytics is a licensed Intune capability; without an entitling plan the node is unavailable or limited. **Verify:** In the Intune admin center, open **Reports** → **Endpoint analytics** and confirm the node loads and is not blocked by a licensing message. **Verify the exact entitling SKUs against current Microsoft documentation** — do not assume a specific bundle grants it. **Implement:** Assign the entitling license (commonly delivered via Microsoft 365 / Intune plans — **confirm the current requirement with Microsoft**) to the devices/users in scope, then re-open the node.

R2. Devices enrolled / co-managed in Intune

Description: The Windows devices you want scored are **Intune-enrolled** or **co-managed** (Configuration Manager + Intune). **Why:** Endpoint Analytics collects its telemetry through the Intune management channel; only managed devices report scores. (Co-managed devices can be onboarded via Configuration Manager — **verify the current co-management onboarding path against Microsoft documentation**.) **Verify:** Intune → **Devices** → **All devices**; the target Windows devices appear and are managed. **Implement:** Enroll the devices in Intune (or onboard the Configuration Manager collection for co-management) before expecting scores. See Global Requirement **G4**.

R3. Supported, current Windows edition/version

Description: A supported Windows edition and version on the devices in scope. **Why:** Startup performance, Work-From-Anywhere, and resource metrics assume a supported, patched Windows build; unsupported builds report partial or no data and drag the WFA score down. **Verify:** Intune → **Devices** → **(device)** → **Hardware** shows the edition/version; or check **Settings** → **System** → **About** on the device. **Confirm the currently supported Windows versions for Endpoint Analytics against Microsoft documentation. Implement:** Bring devices to a supported Windows version with your update policy before baselining.

R4. The Endpoint Analytics data-collection policy

Description: An Endpoint Analytics **data-collection policy** created in Intune and **assigned** to the device group in scope. **Why:** This policy is what tells enrolled devices to **collect and submit** the experience telemetry that produces the scores. No assigned data-collection policy → no data → blank scores. **Verify:** Intune → **Reports** → **Endpoint analytics** → **Settings** (data collection) shows collection configured and assigned to your scope group. **Confirm the current Settings/blade name against Microsoft documentation** (portal wording changes). **Implement:** Create and assign it in Implementation §3 (**named** `EPA – Data Collection – Pilot`), or run `ENDPTAN-Set-DataCollection.ps1` (defaults to `-WhatIf`).

R5. A scoped device group (pilot, then fleet)

Description: A Microsoft Entra security group of the Windows devices to collect from — a small **pilot** group first, then a **fleet** group. **Why:** The data-collection policy is assigned to a group; piloting to a small group first proves data flows and lets you read a baseline before you turn on the whole estate. **Verify:** Entra → **Groups**; `EPA – Pilot Devices` exists with the pilot devices as members. **Implement:** Create it in Implementation §2. (See Global Requirement **G5** for the pilot-group pattern.)

R6. Telemetry / diagnostic data settings permit collection

Description: Windows **diagnostic data** settings on the devices allow Endpoint Analytics to collect what it needs. **Why:** Endpoint Analytics depends on Windows telemetry; if diagnostic data is turned off or set below the level the feature requires, devices will not report complete scores. **Verify:** Check the device's diagnostic-data configuration (via your configuration profiles or **Settings** → **Privacy & security** → **Diagnostics & feedback**). **Confirm the exact diagnostic-data level Endpoint Analytics requires against current Microsoft documentation** — do not hard-code a level. **Implement:** Assign a configuration profile that sets diagnostic data to the level Microsoft currently documents for Endpoint Analytics, to the same scope group.

R7. Network access to the analytics/telemetry endpoints

Description: Outbound HTTPS from managed devices to the Microsoft **Intune** service endpoints **and** the **Windows diagnostic-data / Endpoint Analytics telemetry endpoints**. **Why:** Devices submit their experience telemetry to Microsoft's service; if the telemetry endpoints are blocked at the proxy/firewall, scores stay blank even though the policy is assigned. **Verify:** From a managed device, confirm reachability of the Microsoft telemetry/diagnostic endpoints and the Intune endpoints. **Get the current endpoint list from Microsoft's published documentation** — the diagnostic-data endpoints are a distinct set from the core Intune URLs. **Implement:** Allow the Microsoft-published diagnostic-data + Intune endpoint ranges through proxy/firewall. See Global Requirement **G7**.

R8. Administrative rights to Endpoint Analytics

Description: An operator with rights to read Endpoint Analytics reports and to create/assign the data-collection and configuration policies. **Why:** Enabling data collection and creating policies requires Intune administrative permissions; reading scores requires at least reporting access. **Verify:** Sign in to Intune and confirm you can open **Reports** → **Endpoint analytics** and create policies under **Devices** → **Configuration**. **Implement:** Assign the operator **Intune Administrator** (or a suitable Endpoint Analytics/reporting role — **verify the current role names and scopes against Microsoft documentation**). See Global Requirement **G3**.

R9. (If co-managed) Configuration Manager onboarding for Endpoint Analytics

Description: For **co-managed** devices, the Configuration Manager side is onboarded to Endpoint Analytics. **Why:** Co-managed devices can contribute Endpoint Analytics data via Configuration Manager; this requires the ConfigMgr environment to be connected/onboarded. **Verify:** In the Configuration Manager console, confirm Endpoint Analytics onboarding is configured. **Confirm the current co-management onboarding steps and prerequisites against Microsoft documentation. Implement:** Onboard Configuration Manager to Endpoint Analytics per Microsoft's current guidance (cloud attach / co-management), then confirm the collection is reporting.

R10. National-cloud confirmation (GCC / GCC High / DoD)

Description: If you operate in a **US-sovereign / national cloud**, confirm Endpoint Analytics availability, feature parity, and the correct portal/service endpoints for that cloud. **Why:** Feature availability, rollout timing, and endpoint URLs can differ between Commercial and Government clouds; a Commercial procedure can be wrong for GCC High/DoD. **Verify:** Confirm the Endpoint Analytics feature set and the correct admin portal (e.g. `endpoint.microsoft.us` for some Government tenants) **against Microsoft's US-sovereign-cloud documentation** — do not assume Commercial parity. **Implement:** Use the national-cloud portal and endpoint set Microsoft documents for your cloud; where a capability is delayed or absent, note it and plan around it. **Verify before each deployment.**

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled (or app-managed) devices, regardless of platform. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform and ownership model, then enroll the device.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

G6. Supported, current operating system

Description: A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version or Intune → device → **Hardware**. **Implement:** Deploy update policies or upgrade to a supported release.

G7. Network access to required cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Follow these steps **in order**. The *why* lives in the Solution Overview; the steps below are short and exact so you can **click to implement**. The design is: **enable Endpoint Analytics, assign a named**

data-collection policy to a scoped pilot group, let devices report and set a baseline, interpret each score and its sub-signals, build a prioritized remediation backlog, then expand to the fleet. Pilot first: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Consoles used:

- **Microsoft Intune admin center** — `https://intune.microsoft.com`
- **Microsoft Entra admin center** — `https://entra.microsoft.com`

> ***Portal wording changes.*** *Intune blade and node names (e.g.* ***Reports*** *→* ***Endpoint analytics****, the* ***Settings****/data-collection sub-page, score tab names) are given as they commonly appear, but* ***verify the current navigation against Microsoft documentation*** *before relying on an exact label. The* ***concepts and order*** *below are stable; the* ***exact labels*** *are the part to confirm.*
>
> ***National clouds:*** *if you are in GCC High / DoD, use your cloud's portal (e.g.* `endpoint.microsoft.us`*) and* ***verify feature availability against Microsoft's US-sovereign-cloud documentation*** *(Requirement R10) — the steps are the same in shape but endpoints/parity can differ.*

Phase 1 — Assess & scope

Step 1. Confirm Endpoint Analytics is licensed and the node loads

1. In the Intune admin center, go to **Reports** → **Endpoint analytics**.
2. Confirm the node opens without a licensing/entitlement block. If it is blocked, resolve licensing

(Requirement **R1**) before continuing. **Verify the entitling SKUs against current Microsoft documentation.**

Step 2. Create the pilot device group

1. Entra → **Groups** → **New group** → **Security**; **name it** `EPA – Pilot Devices` (mail nickname

`epa-pilot-devices`).

1. **Membership type = Assigned.** Add a small, representative set of managed Windows devices (mix of

models/roles) as members. *(This is the group the data-collection policy targets first.)*

1. *(Later, in Phase 5, you create `EPA – Fleet Devices` for the whole managed estate.)*

Phase 2 — Configure (enable data collection)

Step 3. Enable Endpoint Analytics and create the data-collection policy

1. In Intune, go to **Reports** → **Endpoint analytics**.
2. Open the **Settings** (data collection) area for Endpoint Analytics. **Confirm the current

Settings/data-collection blade name against Microsoft documentation** — wording varies.

1. **Enable** Endpoint Analytics data collection. When prompted to choose what to collect, enable the

score areas you intend to use (Startup performance, Application reliability, Work-From-Anywhere, Resource performance). **Confirm the current toggle set against Microsoft documentation.**

1. Create/assign the data-collection policy and **name it** `EPA – Data Collection – Pilot`.
2. **Assignments:** assign `EPA – Data Collection – Pilot` to `EPA – Pilot Devices` (Step 2) →

Save.

> *Prefer automation? Run* `ENDPTAN-Set-DataCollection.ps1` *to create/assign the* `EPA – Data Collection – Pilot` *policy to* `EPA – Pilot Devices`*. It* ***defaults to*** `-WhatIf` *(preview only) — re-run without* `-WhatIf` *to apply.* ***Verify the exact Graph resource/settings against current Microsoft documentation****, as the data-collection object can change.*

Step 4. Ensure diagnostic-data settings permit collection

1. Confirm the pilot devices' Windows **diagnostic data** is at the level Endpoint Analytics requires (Requirement **R6**). **Confirm the exact required level against Microsoft documentation.**
1. If needed, create a configuration profile to set it and **name it** `EPA – Diagnostic Data – Pilot`: **Devices** → **Configuration** → **Create** → **New policy** → **Windows 10 and later** → **Settings catalog**; add the diagnostic-data setting Microsoft documents for Endpoint Analytics; **name it as above**; **Assign** to `EPA – Pilot Devices` → **Create**.

Step 5. Confirm the telemetry endpoints are reachable

1. From a pilot device (or via your proxy/firewall team), confirm outbound access to the Microsoft diagnostic-data + Intune endpoints (Requirement **R7**). **Get the current endpoint list from Microsoft's published documentation.**
1. If blocked, allow the published ranges before expecting scores — blocked endpoints are the most common cause of permanently blank scores.

Phase 3 — Pilot (let data flow, set a baseline)

Step 6. Let the pilot devices report

1. Leave the pilot devices powered on and connected through at least one full **collection window**. Scores are **not instant** — allow the time Microsoft documents before expecting populated data. **Confirm the current expected timing against Microsoft documentation.**
1. Force a device sync if you want to hurry policy delivery: **Intune** → **Devices** → **(device)** → **Sync** (this speeds *policy* delivery, not the analytics collection window).

Step 7. Confirm data is flowing

1. Run `ENDPTAN-Test-DataFlow.ps1` to confirm the pilot devices are reporting (read-only), and/or
2. In **Reports** → **Endpoint analytics**, confirm the pilot devices appear and scores begin to populate.

If nothing appears after the documented window, go to Troubleshooting (§6).

Step 8. Establish the pilot baseline

1. In **Reports** → **Endpoint analytics** → **Settings** (baselines), select or confirm the **baseline** to measure against. Start with a **pilot baseline**; you can also compare against the Microsoft-published **"all organizations" median** baseline. **Confirm the current baseline UI/wording against Microsoft documentation.**

1. Record the pilot baseline scores (use `ENDPTAN-Report.ps1`, Step 13) so later improvement is provable.

Phase 4 — Validate (read each score and its sub-signals)

Read each score at a working level. For the **Application/Device reliability** deep-dive (anomaly detection, crash/hang root-cause), use the companion product **Anomaly Detection & Application Reliability with Endpoint Analytics (#3)** — this guide reads the score but does not reproduce that analysis.

Step 9. Read the Startup performance score

1. Open the **Startup performance** score. Note the overall score vs. the baseline.

2. Drill into the **sub-signals** — core boot time, sign-in time, time to a responsive desktop, and the contributing factors (**Group Policy / configuration processing time, startup apps, disk type, drivers/firmware**). **Confirm the current sub-metric names against Microsoft documentation.**

1. Write down the top contributors (e.g. "sign-in dominated by config processing," "spinning disks on model X") — these become backlog items in Step 12.

Step 10. Read the Application reliability score

1. Open the **Application reliability** score. Identify apps with the highest **crash/hang** impact, the devices they affect, and usage counts (to separate broad pain from a one-off).

1. Note the worst offenders for the backlog. *(For a full reliability/anomaly investigation, use companion product #3 — do not attempt the deep root-cause here.)*

Step 11. Read the Work-From-Anywhere and Resource performance scores

1. **Work-From-Anywhere:** open the WFA score and note the readiness sub-signals — **cloud identity (Entra-joined), cloud management (Intune/co-managed), Windows edition/version currency, Autopilot, and TPM/hardware readiness**. Note which factor is dragging the score down.

1. **Resource performance:** open the Resource performance score and note **CPU** and **RAM** pressure — flag under-spec'd or over-loaded devices.

1. **Confirm the current sub-signal names/thresholds against Microsoft documentation.**

Phase 5 — Build the backlog, then expand

Step 12. Build the prioritized remediation backlog

1. From the sub-signals in Steps 9–11, assemble a single **prioritized backlog** — highest-impact, lowest-effort first. Group items by lever:

- **Policy/config** (e.g. reduce Group Policy processing time, trim startup apps).
- **Apps** (address the worst crash/hang apps — hand to reliability deep-dive #3 if needed).
- **Drivers/firmware** (update outdated drivers dragging startup).
- **Hardware/edition** (disk upgrades, RAM, Windows edition currency for WFA).

1. For items you will fix **at scale with detection/remediation scripts**, hand them to the companion product **Proactive Remediations at Scale — Detection & Remediation Script Packs for Microsoft Intune (#2)**, which is the engine for authoring and deploying those script pairs. *(This guide builds the backlog; #2 builds and runs the remediation script packs.)*

1. Record the backlog with each item's score area, contributing sub-signal, target, and owner.

Step 13. Report and record the baseline + backlog

1. Run `ENDPTAN-Report.ps1` to export a CSV summary of the scores/sub-signals for your records and for stakeholder reporting. Keep it alongside the pilot baseline from Step 8.

Step 14. Create the fleet group and expand data collection

1. Entra → **Groups** → **New group** → **Security**; **name it** `EPA – Fleet Devices` (mail nickname `epa-fleet-devices`). Add the full managed Windows estate (or use a dynamic device group keyed on your management/OS criteria).

1. In **Reports** → **Endpoint analytics** data-collection settings, **add** `EPA – Fleet Devices` to the assignment of the data-collection policy (rename the policy to `EPA – Data Collection` if you prefer a fleet-wide name, and update the diagnostic-data profile assignment to match).

1. Let the fleet report through a collection window, then establish the **fleet baseline** (Step 8, applied fleet-wide).

Step 15. Operationalize the improvement loop

1. Set a **review cadence** (e.g. monthly): re-read each score against the baseline, refresh the backlog, and confirm remediated items moved the score.

1. Feed backlog items to **Proactive Remediations (#2)** for scaled fixes and to the **reliability/anomaly deep-dive (#3)** for app reliability investigations.

1. Re-run `ENDPTAN-Report.ps1` each cycle to show the trend — the delta from baseline is your proof of improvement.

5. Testing & Validation — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Validate against the **pilot device group** before expanding to the fleet. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts (Microsoft Graph) for tenant-side checks and the Intune portal for the score/baseline checks.

> ***Scores are not instant.*** *Several tests below depend on the Endpoint Analytics* ***collection window*** *— allow the time Microsoft documents after enabling data collection before treating a blank result as a failure.* ***Confirm the current expected timing against Microsoft documentation.***

Test 1 — Tenant readiness

Procedure: Run `ENDPTAN-Discovery.ps1`. **Pass:** Endpoint Analytics data collection is configured, the `EPA – Pilot Devices` group exists, and managed Windows devices are present in the tenant. **Fail:** Any missing → resolve before baselining (create the group, enable data collection, enroll devices).

Test 2 — Data-collection policy assigned to the pilot

Procedure: In Intune → **Reports** → **Endpoint analytics** → **Settings** (data collection), confirm `EPA – Data Collection – Pilot` is enabled and assigned to `EPA – Pilot Devices`. **Verify the current Settings wording against Microsoft documentation. Pass:** The data-collection policy is **enabled and assigned** to the pilot group. **Fail:** Not assigned → assign it (Implementation §3) or run `ENDPTAN-Set-DataCollection.ps1 -Apply`.

Test 3 — Data is flowing from the pilot devices

Procedure: Run `ENDPTAN-Test-DataFlow.ps1` (and confirm the pilot devices appear in **Reports** → **Endpoint analytics**). **Pass:** The pilot devices are **managed, syncing recently, and reporting** analytics data (or the documented collection window has not yet elapsed and they are otherwise healthy). **Fail:** Devices not reporting after the documented window → see Troubleshooting (no data / blocked endpoints / diagnostic-data level / licensing).

Test 4 — Scores populated for the pilot

Procedure: In **Reports** → **Endpoint analytics**, confirm the pilot devices have populated scores for the enabled areas (Startup performance, Application reliability, Work-From-Anywhere, Resource performance). **Pass:** Scores are **populated** (not blank/"collecting") for the pilot devices. **Fail:** Blank after the documented window → Troubleshooting (scores blank or delayed).

Test 5 — Baseline established

Procedure: In **Reports** → **Endpoint analytics** → **Settings** (baselines), confirm a **baseline** is selected/established, and record the pilot baseline scores (use `ENDPTAN-Report.ps1`). **Pass:** A **pilot baseline** exists and its scores are recorded for later comparison. **Fail:** No baseline → establish one (Implementation §8); **verify the baseline UI against Microsoft documentation**.

Test 6 — A score sub-signal is correctly interpreted

Procedure: Pick one below-baseline score (e.g. Startup performance) and drill into its sub-signals; identify the top contributing factor (e.g. Group Policy / configuration processing time, a startup app, disk type, or an

outdated driver) and record it as a backlog item. **Pass:** The low score is traced to a **specific, named sub-signal** and captured as a prioritized backlog item (not just "the score is low"). **Fail:** You can see the score but not the cause → re-check that the relevant score area was enabled in data collection (Implementation §3) and that the sub-signals have populated.

Test 7 — Backlog produced and reportable

Procedure: Run `ENDPTAN-Report.ps1` and confirm you have a CSV of scores/sub-signals plus a written prioritized backlog (Implementation §12). **Pass:** A prioritized remediation backlog exists, each item tied to a score area and sub-signal, with a report/export saved alongside the baseline. **Fail:** No actionable backlog → complete Implementation §9–§12.

Acceptance gate

Proceed to **Expand** (fleet) only when Tests 1–7 pass: Endpoint Analytics is **enabled**, the **data-collection policy is assigned** to the pilot, **data is flowing**, **scores are populated**, a **baseline is established**, at least one **sub-signal is correctly interpreted**, and a **prioritized remediation backlog** exists. Hand backlog items to **Proactive Remediations (#2)** for scaled fixes and to the **Anomaly Detection & Application Reliability product (#3)** for reliability deep-dives.

6. Troubleshooting — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Work top-down: confirm **licensing/access**, then **data collection is enabled and assigned**, then **diagnostic-data settings**, then **endpoint reachability**, then **scope**, then give it the **collection window**. Most "no scores" cases are one of: data collection not assigned, diagnostic data below the required level, blocked telemetry endpoints, or simply not enough time elapsed yet.

> ***Before chasing a fault, remember scores are not instant.*** *After enabling data collection there is a* ***collection window*** *before scores populate and a baseline forms.* ***Confirm the current expected timing against Microsoft documentation*** *and wait it out before treating blank scores as broken.*

No data at all / devices not reporting

- **Data-collection policy not assigned.** In **Reports** → **Endpoint analytics** → **Settings**, confirm `EPA – Data Collection – Pilot` is **enabled and assigned** to `EPA – Pilot Devices` (Implementation §3). Run `ENDPTAN-Test-DataFlow.ps1` to see which pilot devices are reporting.
- **Diagnostic data below the required level.** Endpoint Analytics needs Windows diagnostic data at the level Microsoft documents; if it's off or too low the device won't report. Confirm/assign `EPA – Diagnostic Data – Pilot` (Implementation §4). **Verify the required level against Microsoft documentation** — do not assume a fixed level.
- **Telemetry endpoints blocked.** The diagnostic-data / analytics endpoints are a **distinct set** from the core Intune URLs; if the proxy/firewall blocks them, scores stay blank even with the policy assigned. Allow the **Microsoft-published** endpoint ranges (Requirement R7) and re-test.

- **Device not actually managed / not syncing.** Confirm the device is Intune-enrolled or co-managed and

syncing (Intune → Devices → device → **Sync**). A device that isn't checking in can't report analytics.

Scores blank or delayed (data is flowing but no scores)

- **Collection window not elapsed.** Give it the documented time; `ENDPTAN-Test-DataFlow.ps1` can show

devices are syncing even before scores render.

- **Score area not enabled.** If a specific score is missing, confirm that area was enabled when you

turned on data collection (Implementation §3) — e.g. Application reliability enabled for the app-health score. **Verify the current toggle set against Microsoft documentation.**

- **Too few devices for a meaningful score.** Some scores need a minimum population/usage before they

render. Expand the pilot slightly or wait for the fleet. **Confirm any current minimums with Microsoft.**

- **Co-managed devices not onboarded.** For co-managed devices, confirm the Configuration Manager side is

onboarded to Endpoint Analytics (Requirement R9). **Verify the current onboarding steps with Microsoft.**

Licensing / prerequisite gaps

- **Node blocked or limited.** If **Reports** → **Endpoint analytics** shows a licensing block, resolve the

entitlement (Requirement R1) — **verify the entitling SKUs against current Microsoft documentation**; don't assume a bundle includes it.

- **Insufficient rights.** If you can read but not create the data-collection/config policies, you likely

lack Intune administrative rights (Requirement R8) — assign the appropriate role.

- **Unsupported Windows version.** Devices on an unsupported build report partial/no data and drag WFA

down. Bring them current (Requirement R3).

Scope wrong (right feature, wrong devices)

- **Policy assigned to the wrong group.** Confirm the data-collection policy targets `EPA – Pilot Devices`

(pilot) or `EPA – Fleet Devices` (fleet) as intended, and that the diagnostic-data profile is assigned to the **same** group — a mismatch means devices collect but aren't told to, or vice versa.

- **Devices missing from the group.** Confirm the intended devices are members (Assigned group) or match

the rule (dynamic group). Re-run `ENDPTAN-Validate.ps1` after fixing membership.

- **Fleet expansion missed the diagnostic-data profile.** When you add `EPA – Fleet Devices` to data

collection (Implementation §14), also extend the `EPA – Diagnostic Data` profile to the fleet group.

National-cloud differences (GCC / GCC High / DoD)

- **Wrong portal or endpoint set.** Government tenants may use a different admin portal

(e.g. `endpoint.microsoft.us`) and different service/telemetry endpoints, and a feature may be delayed or absent. **Verify availability, the correct portal, and the endpoint list against Microsoft's US-sovereign-cloud documentation** before concluding something is broken.

Scripts (PowerShell / Microsoft Graph)

- **Connect with the right scopes.** The read-only scripts use read scopes

(`DeviceManagementConfiguration.Read.All`, `DeviceManagementManagedDevices.Read.All`, `DeviceManagementServiceConfig.Read.All`, `Group.Read.All`); the state-changer (`ENDPTAN-Set-DataCollection.ps1`) needs `DeviceManagementConfiguration.ReadWrite.All` and `Group.ReadWrite.All`. Consent to exactly those scopes at `Connect-MgGraph` — a missing scope shows as a 403/authorization error.

- **The state-changer defaults to `-WhatIf`.** `ENDPTAN-Set-DataCollection.ps1` **previews only** unless

you pass `-Apply`. If "nothing changed," that's expected — re-run with `-Apply` to commit.

- **Match names/scope exactly (verbatim).** The scripts look up objects by their exact display names

(`EPA – Pilot Devices`, `EPA – Data Collection – Pilot`). If you renamed an object, pass the matching `-PilotGroupName` / `-PolicyName` so the lookup finds it — note the names use an en dash (–), not a hyphen; copy them exactly.

- **Graph resources evolve — verify.** The Endpoint Analytics `userExperienceAnalytics*` resources are on

the **beta** endpoint and change over time. If `ENDPTAN-Report.ps1` / `ENDPTAN-Test-DataFlow.ps1` return nothing, read the scores in **Reports** → **Endpoint analytics** and **verify the current Graph resource names against Microsoft documentation** before assuming a fault. The scripts intentionally do **not invent** an endpoint — `ENDPTAN-Set-DataCollection.ps1` previews its intent and points you to the portal when the data-collection resource isn't set.

- **Module missing.** `Install-Module Microsoft.Graph -Scope CurrentUser` once before first run.

Useful checkpoints

- **Intune: Reports** → **Endpoint analytics** (scores/baseline), **Reports → Endpoint analytics →

Settings **(data collection + baseline),** Devices → Configuration **(the diagnostic-data profile),** Devices → All devices** (managed/sync state).

- **Entra: Groups** → `EPA – Pilot Devices` **/** `EPA – Fleet Devices` (membership).

- **Scripts:** re-run `ENDPTAN-Validate.ps1` after each change to confirm the gate.

7. Best Practices — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Field-proven guidance for standing up Endpoint Analytics and running it as an ongoing improvement loop.

Deploy & scope

- **Pilot before fleet, always.** Assign the `EPA – Data Collection – Pilot` policy to a small, representative `EPA – Pilot Devices` group first — prove data flows and read a baseline before you turn on the whole estate.
- **Make the pilot representative.** Include a mix of models, roles, and locations so the pilot baseline reflects reality, not just the newest laptops.
- **Name everything by intent** — `EPA – Pilot Devices`, `EPA – Data Collection – Pilot`, `EPA – Diagnostic Data – Pilot`, `EPA – Fleet Devices` — so the configuration is self-documenting and easy to audit and clean up.
- **Confirm diagnostic-data + endpoints up front.** The two most common causes of blank scores are diagnostic data set below the required level and blocked telemetry endpoints — verify both before you wait on a collection window. **Confirm the required level and endpoint list against Microsoft docs.**

Baseline & measure

- **Set a baseline before you change anything.** A score is only meaningful against a baseline — capture the pilot baseline (and the fleet baseline later) and record the numbers so improvement is provable.
- **Compare against the "all organizations" median too.** Your own baseline shows movement; the Microsoft-published median shows whether you are ahead of or behind the pack.
- **Expect a collection window.** Scores populate after a delay, not instantly — don't declare failure on day one. **Verify the current timing against Microsoft documentation.**

Interpret the scores

- **Always drill from score to sub-signal.** "Startup is low" is not actionable; "sign-in is dominated by Group Policy processing on model X" is. Trace every low score to a named cause.
- **Separate broad pain from one-offs.** For Application reliability, weigh crash/hang counts against usage — a widely used app failing often beats a single flaky device. For the deep reliability + anomaly investigation, use the companion product **Anomaly Detection & Application Reliability with Endpoint Analytics (#3)**.

• **Treat Work-From-Anywhere as a modernization checklist.** Its sub-signals (cloud identity, cloud management, edition currency, Autopilot, TPM) each map to a concrete project — attack them one at a time.

Remediate & operate

• **Turn scores into a ranked backlog, not a data dump.** Prioritize highest-impact, lowest-effort first, and tie every item to a score area and sub-signal.

• **Fix at scale with the remediation engine.** For anything you'd script (detection + fix), use the companion product **Proactive Remediations at Scale — Detection & Remediation Script Packs for Microsoft Intune (#2)** rather than one-off manual fixes.

• **Re-measure against the baseline after every change.** The delta from baseline is your proof the work paid off — keep it for stakeholder reporting.

• **Run a review cadence.** Re-read the scores on a schedule (e.g. monthly), refresh the backlog, and report the trend. `ENDPTAN-Report.ps1` gives you the export each cycle.

• **Keep a runbook** of the groups, the data-collection policy, the diagnostic-data profile, the baseline(s), and the backlog — the moving parts that define your Endpoint Analytics program.

National clouds

• **Verify before you deploy in GCC High / DoD.** Feature availability, rollout timing, and portal endpoints can differ from Commercial — confirm against Microsoft's US-sovereign-cloud documentation and use the correct national-cloud portal.

8. Deliverables Summary — Endpoint Analytics Deployment & Optimization in Microsoft Intune

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

• **Executive Summary** — the business case and outcome in one page.

• **Solution Overview** — what Endpoint Analytics is, the four score areas and what each measures, how baselining/benchmarking works, the improvement loop, where it fits in the three-product cluster (pointers to the Proactive Remediations and reliability/anomaly siblings), benefits, and honest limitations.

• **Requirements Guide** — every prerequisite (Endpoint Analytics licensing, enrolled/co-managed devices, supported Windows, the data-collection policy, a scoped device group, diagnostic-data settings, telemetry endpoints, admin rights, co-management onboarding, national-cloud confirmation) with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.

• **Implementation Guide** — exact, ordered, click-by-click steps across five phases: assess & scope, configure (enable data collection), pilot (let data flow, set a baseline), validate (read each score and its sub-signals), then build the backlog and expand to the fleet — every object named and reused.

• **Testing & Validation** — seven tests with explicit pass/fail gates and an acceptance gate, proving data flows, scores populate, a baseline is set, and a sub-signal is correctly interpreted into a backlog.

• **Troubleshooting** — top-down diagnosis for no data, blank/delayed scores, licensing/prereq gaps, wrong scope, and a scripts section (Graph scopes, exact-name matching, `-WhatIf` default).

• **Best Practices** — deploy/scope, baseline/measure, interpret, remediate/operate, and national clouds.

PowerShell scripts (Microsoft Graph) — `ENDPTAN-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK); each has a header explaining **Purpose · How to run · Graph scopes · Expected results**. Four are **read-only**; one is a **state-changer that defaults to** `-WhatIf`.

1. `ENDPTAN-Discovery.ps1` — read-only tenant readiness: Endpoint Analytics data collection present, the pilot group exists, and managed Windows devices in scope.
1. `ENDPTAN-Validate.ps1` — PASS/FAIL gate: data collection configured, pilot group populated, and pilot devices managed and syncing recently.
1. `ENDPTAN-Report.ps1` — exports a CSV summary of the scores and sub-signals for records, baseline tracking, and stakeholder reporting.
1. `ENDPTAN-Test-DataFlow.ps1` — read-only confirmation that the pilot devices are submitting analytics data, calling out any device not yet reporting.
1. `ENDPTAN-Set-DataCollection.ps1` — the only state-changer: creates/assigns the `EPA – Data Collection – Pilot` policy to `EPA – Pilot Devices`. **Defaults to** `-WhatIf` (preview only); re-run with `-Apply` to commit.

> *The Endpoint Analytics Graph resources evolve; the scripts flag where to **verify the current resource names against Microsoft documentation** and fall back to the portal where Graph is inconclusive.*

Hero image

• `ENDPTAN-hero.png` — the product image (1200×1200) used on the marketplace listing.

How it's delivered

A **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

Companion products (buy for depth — not reproduced here)

- **Proactive Remediations at Scale — Detection & Remediation Script Packs for Microsoft Intune (#2)** —

the engine that fixes backlog items at scale (authoring/packaging/scheduling detection + remediation scripts).

- **Anomaly Detection & Application Reliability with Endpoint Analytics (#3)** — the reliability and anomaly deep-dive (crash/hang root-cause, catching bad rollouts early).

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ENDPTAN-Discovery.ps1

Purpose: Confirms the building blocks of an Endpoint Analytics rollout exist: an Endpoint Analytics data-collection setting/policy is present, the pilot device group exists and is assigned, which managed Windows devices are in scope, and (best-effort) which devices are already reporting. Reports PASS/NOTE per check with an overall verdict.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ENDPTAN-Discovery.ps1 .\ENDPTAN-Discovery.ps1 -PilotGroupName 'EPA - Pilot Devices' Reads only — makes no changes.

ENDPTAN-Report.ps1

Purpose: Pulls the Endpoint Analytics scores and their key sub-signals (Startup performance, Application reliability, Work-From-Anywhere, Resource performance) and writes a CSV summary for records, baseline tracking, and stakeholder reporting. Best-effort per-device pull where Graph exposes it, plus an overall-score summary.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ENDPTAN-Report.ps1 .\ENDPTAN-Report.ps1 -CsvPath C:\Temp\endpoint-analytics.csv

ENDPTAN-Set-DataCollection.ps1

Purpose: The ONE state-changing script in this pack. Ensures the pilot device group exists, then creates/assigns the Endpoint Analytics data-collection policy ('EPA – Data Collection – Pilot') to that group so pilot devices begin submitting experience telemetry. Idempotent — skips objects that already exist.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ENDPTAN-Set-DataCollection.ps1 # preview only (default, no changes) .\ENDPTAN-Set-DataCollection.ps1 -Apply # actually create/assign

ENDPTAN-Test-DataFlow.ps1

Purpose: For the pilot device group, confirms (read-only) that the members are managed Windows devices checking in recently and, best-effort, appearing in the Endpoint Analytics device-scores data — i.e. telemetry is flowing and scores are populating. Lists any pilot device that is NOT yet reporting so you know what to chase.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ENDPTAN-Test-DataFlow.ps1 .\ENDPTAN-Test-DataFlow.ps1 -PilotGroupName 'EPA - Pilot Devices' -RecentDays 3

ENDPTAN-Validate.ps1

Purpose: Validates the acceptance gate for the Endpoint Analytics pilot: Endpoint Analytics data collection is configured, the pilot group exists and has members, the pilot devices are managed Windows devices reporting recently, and (best-effort) scores/baseline are available. Produces a clear PASS/FAIL verdict.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ENDPTAN-Validate.ps1 .\ENDPTAN-Validate.ps1 -PilotGroupName 'EPA - Pilot Devices' -RecentDays 7